Pradosh Barun Das

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Academic Education

Ph.D. in Astrophysics & Astronomy

Macquarie University, Sydney, Australia

Awarded the "Australian International Research Training Program (iRTP) Scholarship" *April 2023 - Ongoing*

PhD Thesis Title: Data-Driven Galactic Archaeology: Tracing the Chemodynamical Structure and Accretion History of the Milky Way in an Era of Large Spectroscopic Surveys

Supervisors: Prof. Daniel Zucker (MQ), A/Prof. Gayandhi De Silva (MQ) & Francesca Primas (ESO Garching)

Research Interests: Galactic Archaeology, Milky Way Formation and Evolution, Chemical Tagging of accreted Stellar Populations

Bachelor & Master of Science (BS-MS Dual Degree) (Major in Physics, Specialization in Astronomy)

Indian Institute of Science Education and Research (IISER) Mohali, India

Awarded the "DST-INSPIRE Undergraduate Scholarship" (Department of Science & Technology, India) *Aug 2017 - May 2022*

Master's Thesis: Model Independent Analysis of Cosmological Parameters

Supervisor: A/Prof. Harvinder Kaur Jassal

CPI (Cumulative Performance Index): **9.0/10.0**

Research Experience

1. Barium & Europium abundances from the *Gaia*-ESO survey using *The Cannon*

ESO Garching, Germany

Dr Francesca Primas, A/Prof. Gayandhi De Silva, Dr Guillaume Guiglion, Prof. Daniel Zucker and Prof. Melissa Ness

ESO Research Project 2025

- Applied ***The Cannon*** to determine neutron-capture elemental abundances (Ba & Eu) and stellar parameters for ∼40,000 *Gaia*-ESO FLAMES/GIRAFFE HR15N spectra.
- Derived new Ba abundances for ∼10,000 stars and new Eu abundances for ∼100 stars, substantially expanding the neutron-capture element catalogue beyond previous *Gaia*-ESO data releases.
- The developed framework can be readily adapted to analyse spectra from upcoming 4MOST, WEAVE, and *Gaia* DR4 (RVS) surveys.

2. The GALAH Survey: Neutron-Capture Elemental Abundances for 350,000 *Gaia* RVS Spectra & the Chemodynamics of Accreted Structures

Macquarie University, Sydney, Australia

Prof. Daniel Zucker, A/Prof. Gayandhi De Silva, and The GALAH Collaboration

Ph.D. Research Project

- Determined neutron-capture elemental abundances (Nd, Ce, Zr) for ∼350,000 *Gaia* DR3 RVS giant stars using high-resolution labels from GALAH DR4.
- Performed chemo-dynamical tagging to identify potential accreted stellar members associated with the *Gaia*-Sausage-Enceladus (GSE) accreted structure in the Galactic halo by developing a probabilistic MCMC model using only chemical abundances – Future work will extend this analysis to other accreted populations, including Sequoia and Thamnos.
- Key results included finding the combined [n-capture/α] elemental ratios which distinguish GSE structure members using chemical information alone.
- Demonstrated the ability to recover neutron-capture chemical signatures from low-resolution *Gaia* RVS spectra, paving the way for future population studies of accreted structures in the Milky Way.

3. The GALAH Survey: Stellar Parameters and Abundances for 800,000 *Gaia* RVS Spectra using GALAH DR4 & *The Cannon*

Macquarie University, Sydney, Australia

Prof. Daniel Zucker, A/Prof. Gayandhi De Silva, and The GALAH Collaboration — *Ph.D. Research Project*

- Applied ***The Cannon*** to infer $T_{\rm eff}$, $\log g$, $v \sin i$, [Fe/H], and elemental abundances (Ca, Si, Ni, Ti) for ∼800,000 *Gaia* DR3 RVS spectra ($R \sim 11,500$) by training on ∼14,500 *Gaia* RVS spectra with stellar labels from high-resolution GALAH DR4 ($R \sim 28,000$), achieving high-precision.
- One of the first studies to detect a clear [Ti/Fe]-[Fe/H] bimodality using *Gaia* DR3 RVS, enabling large-scale, scalable stellar population analysis across the Milky Way.

4. Kinematics & Spatial Structure of YSOs in the Upper Scorpius

Chalmers University of Technology, Sweden

Prof. Jonathan Tan and Dr. Joseph Armstrong — *June – August 2022*

- Analyzed spatial and kinematic structures of ∼400 Young Stellar Objects (YSOs) in Upper Scorpius, including 250 identified via the *Gaia* DR3 Photometric Variability catalogue.
- Identified several dense stellar groups with distinct motions contributing to the formation of the Upper Scorpius OB association.

5. Model-Independent Analysis of Cosmological Parameters

IISER Mohali, India

Assoc/Prof. Harvinder Kaur Jassal — *May 2021 – May 2022*

- Applied non-parametric Gaussian Process (GP) regression model on Hubble parameter and Pantheon supernova datasets.
- Explored effects of kernel choices (RBF, Matérn 3/2 and 5/2) and error models on cosmological inference.

6. Repeater vs. Non-Repeater Fast Radio Bursts (FRBs)

National Centre for Radio Astrophysics, TIFR, Pune, India

Dr. Jayanta Roy — *May – July 2021*

- Analyzed CHIME-FRB Survey Catalogue-1 & FRBCAT Transient Server Catalogue to identify differences in properties between repeating & non-repeating FRBs.
- Examined frequency occupancy across CHIME and GMRT datasets for selected repeater events.

7. High-Resolution Spectroscopy of BD-19132: A Carbon-Enhanced Metal-Poor Star

Indian Institute of Astrophysics, Bengaluru, India

Prof. Aruna Goswami — *May – July 2019*

- Performed detailed spectral data reduction procedure followed by analysing properties of rapid(r-) and slow(s-) process nucleosynthesis channels.
- Found and classified BD-19132 as a Carbon-Enhanced Metal-Poor (CEMP) star enhanced by both r- and s-processes using the reduced HESP spectra from the Himalayan Chandra Telescope.
- Measured photometric temperature and radial velocity to support abundance analysis.

Fellowships, Grants & Awards

- **ORIGINS Excellence Cluster Scientific Visitorship** (2026):
  Selected for scientific visitorship at the European Southern Observatory (ESO), Garching, Germany to work with Dr. Francesca Primas and Dr. Amelia Bayo on galactic archaeology using ESO archival spectroscopy and Gaia stellar catalogues and develop a scientific synergy with forthcoming large-scale spectroscopic surveys and facilities such as 4MOST & MOONS.
- **ESO Early-Career Scientific Research Visitorship** (2025):
  Selected for Early-career scientific research visit to the European Southern Observatory (ESO), Garching, Germany to work with Dr. Francesca Primas on Galactic Archaeology using Data-driven models.
- **Post-Graduate Research Fund (PGRF) Grant**, Macquarie University (2025):
  *Awarded a competitive research grant of **AUD 6000** by Macquarie University, Sydney, Australia*
- **ASA Student Travel Assistance Scheme (ASA-STAS) Grant** (2025):
  *Awarded a competitive grant of **AUD 1500** by the Astronomical Society of Australia (ASA)*
- **Graduate Research Excellence Award, Macquarie University** (2025):

Nominated for the award by the School of Mathematical and Physical Sciences (MAPS), recognizing excellence in academic and research performance

- **International Research Program (RTP) Scholarship** (2023):
  Awarded the competitive Australian Commonwealth Government funded RTP Scholarship for PhD

- **Chalmers Astrophysics and Space Sciences Summer Fellowship (CASSUM)** (2022):
  *Awarded a competitive fellowship of **25,000 SEK** jointly by the Chalmers University of Technology, Sweden and the University of Virginia, USA to work on Young stellar associations*

- **VSRP Research Fellowship**, National Centre for Radio Astrophysics (NCRA) – TIFR (2021):
  *Selected **among 4 students nationally** in India to work on a research project at the NCRA on Fast Radio Bursts. Awarded a scholarship of **Rs. 30,000** under the programme*

- **IASc–INSA–NASI Summer Research Fellowship** (2019):
  *Awarded jointly by the Indian Academy of Sciences (IASc), the Indian National Science Academy (INSA), and the National Academy of Sciences (NASI), to conduct research on Carbon-Enhanced Metal Poor Stars using High-resolution spectroscopy at the Indian Institute of Astrophysics, Bengaluru, India. Scholarship: **Rs. 30,000***

- **NIUS Physics & Astronomy Camp, HBCSE–TIFR** (2018):
  *Selected **nationally in India on merit-based** to attend Homi Bhabha Centre for Science Education (HBCSE) - National Undergraduate Physics & Astronomy camp (NIUS) at TIFR, Mumbai, India.*

- **DST-INSPIRE Scholarship** (2017–2022):
  *Awarded the national undergraduate merit scholarship (INSPIRE) from Department of Science & Technology (DST), India. Awarded **Rs 360,000** under the Scholarship*

- **International Olympiads & Contests (Mathematics)**:
  - **IMO – International Mathematics Olympiad (SOF)**:
    - 2015 – State rank **1 (Gold)**, International rank **23**
    - 2016 – Zonal rank **9**, International rank **381**
  - **IOM – International Olympiad of Mathematics (Silverzone)**:
    - 2015 – State rank **2 (Silver)**, International rank **85**
    - 2015 – State rank **1 (Gold)**, International rank **25**
    - 2016 – State rank **1 (Gold)**, International rank **179**

Conferences, Posters & Seminars

- **Contributed Talk (2026)**, "Charting the Future of Stellar and Exoplanet Spectroscopy (CHASES 2026) Conference", Jointly organised by INAF, OCA and ESO Garching at Elba Island (Italy)
- **Invited Talk (2026)**, "Australian National University Seminar Series", Canberra (Australia)
- **Contributed Talk (2026)**, "Southern Stellar Stream Spectroscopic Survey (S5) Science Meeting", Macquarie University (Australia)
- **Invited Seminar (2025)**, "Max Planck Institute for Astronomy-Seminar", Heidelberg (Germany)
- **Invited Seminar (2025)**, "University of Ljubljana-Astrodebata, FMF", Ljubljana (Slovenia)
- **ESO-ECR Seminar (2025)**, "European Southern Observatory-Garching", (Germany)
- **Short talk** & **Poster (2025)**, "European Astronomical Society Annual Meeting", Cork (Ireland)
- **Contributed talk (2025)**, "Astronomical Society of Australia (ASA) Annual Science Meeting", University of Adelaide (Australia)
- **Contributed talk and Poster (2025)**, "Australian Institute of Physics (AIP)-Macquarie University", Sydney (Australia)
- **Contributed talk (2024)**, "Stars in Brisbane & Australian Exoplanet Workshop X", University of Southern Queensland, Brisbane (Australia)
- **Contributed talk (2024)**, "Astronomical Society of Australia (ASA) Annual Science Meeting", CSIRO, Sydney (Australia)

- **Contributed talk (2024)**, "ASTRO 3D Science Legacy Meeting", Sydney (Australia)
- **Contributed talk (2023)**, "American Astronomical Society (AAS)", Seattle, Washington (USA)
- **Contributed talk (2022)**, "From Stars to Galaxies II (FSTGII)", Chalmers University of Technology - Gothenburg (Sweden)
- **Contributed talk (2021)**, "VSRP Seminar", National Centre for Radio Astrophysics, TIFR (India)

Scientific Outreach

- **Scientific Assistant** (2025), ESO OPC Expert Panels: Supporting the ESO Observing Programmes Committee for reviewing telescope observing proposals.
- **Student Representative** (2023–ongoing), Astronomical Society of Australia (ASA): Represent the astronomy student research cohort nationwide for the Astronomical Society of Australia (ASA) National Council, contributing to policy and event planning for the student community.
- **President** (2025–2026), Physics and Astronomical Society (MUPAS), Macquarie University, Australia: Led a society of 300+ students from diverse disciplines to promote inclusion, diversity, equity, and accessibility in physics- and astronomy-focused events, workshops, and student-led research discussions.
- **Lead Organiser** (2024–25), Astronomy Seminar Series, Macquarie University, Australia: Coordinated a seminar series featuring international and domestic speakers, fostering research discussions and collaborative opportunities.
- **Lead Organiser** (2024), Harley Wood School of Astronomy (HWSA, ASA), Australia: Organized a 3-day national student astronomy retreat including invited research talks, workshops, and hands-on sessions. The event welcomed Astronomy researchers from all across Australia.
- **Observatory Supervisor** (2023–ongoing), Association for Astronomy (AfA), Macquarie University Observatory, Australia: Trained and supervised undergraduate students in telescope operations and organized public outreach events.
- **Teaching Associate** (2023–ongoing), Macquarie University, Australia: Supervised undergraduate students in Physics & Astronomy academic tutorials and laboratory sessions.

Publications/Manuscripts

- **Das, P. B.**, Zucker, D. B., De Silva, G. M., Borsato, N. W., Mura-Guzmán, A., Buder, S., Ness, M., Nordlander, T., Casey, A. R., Martell, S. L., Bland-Hawthorn, J., de Grijs, R., Freeman, K. C., Kos, J., Stello, D., Lewis, G. F., Hayden, M. R., & Sharma, S. (2025). *The GALAH Survey: Stellar parameters and abundances for 800,000 Gaia RVS spectra using GALAH DR4 and The Cannon. Monthly Notices of the Royal Astronomical Society*, **538**(2), 605–621. https://doi.org/10.1093/mnras/staf169
- **Das, P. B.**, Zucker, D. B., Mura-Guzmán, A., Borsato, N. W., De Silva, G. M., Buder, S., Feuillet, D., Nordlander, T., Ness, M., Martell, S. L., Kos, J., de Grijs, R., Bland-Hawthorn, J., Freeman, K. C., Casey, A. R., Lewis, G. F., Stello, D., & the GALAH Collaboration (2026). *The GALAH Survey: Neutron-Capture Elemental Abundances for 350,000 Gaia-RVS Spectra and the Chemodynamics of Accreted Structures. Monthly Notices of the Royal Astronomical Society*, **550**(4), stag1142. https://doi.org/10.1093/mnras/stag1142
- **Das, P. B.**, Primas, F., De Silva, G. M., Guiglion, G., Zucker, D. B., & Ness, M. (2026). *Barium and europium abundances from the Gaia-ESO Survey using The Cannon. Astronomy & Astrophysics*, **713**, A148. https://doi.org/10.1051/0004-6361/202660092
- Wang, E. X., Ness, M., Nordlander, T., Casey, A. R., Martell, S. L., Pinsonneault, M., Fu, X., Stello, D., Reyes, C., Hon, M., McKenzie, M., Jian, M., Yu, J., Buder, S., Lind, K., Bland-Hawthorn, J.,

Zucker, D. B., **Das, P. B.**, de Grijs, R., & Hayden, M. R. (2025). *Optical spectroscopic signatures of the red giant evolutionary state. Monthly Notices of the Royal Astronomical Society*, **540**(4), 3919–3933. https://doi.org/10.1093/mnras/staf916

- **Das, P. B.** et al. (2025). *Gaia RVS stellar labels using The Cannon.* CDS/VizieR Catalogue, **J/MNRAS/538/605**. https://vizier.cds.unistra.fr/viz-bin/cat/J/MNRAS/538/605
- **Das, P. B.** et al. (2026). *Gaia-ESO HR15N Ba and Eu abundances.* CDS/VizieR Catalogue, **J/A+A/713/A148**. https://vizier.cds.unistra.fr/viz-bin/cat/J/A+A/713/A148
- **Das, P. B.**, Armstrong, J., & Tan, J. (2023). *The structure and kinematics of the newborn stars of Upper Scorpius. Bulletin of the American Astronomical Society*, **55**(2), AAS241. 2023AAS...24133406D
- **Das, P. B.**, & Roy, J. (2021). *Characterizing Fast Radio Bursts: Repeater vs Non-Repeater Properties and Telescope Biases.* NCRA–TIFR VSRP Fellowship Research Report.
- **Das, P. B.**, & Goswami, A. (2019). *Understanding the Spectral Characteristics of CEMP Star BD-19132, using High-Resolution Spectroscopy.* IASc–INSA–NASI Summer Research Fellowship Report.

Observing Proposals and Experience

- Semester 2026B | R/2026B/05 | GALAH 2 Key Program| Anglo-Australian Telescope | Co-Investigator / Observer | Australian Astronomical Observatory, Siding Spring Observatory - *Veloce*
- Semester 2025B | R/2025B/01 | GALAH 2 | Anglo-Australian Telescope | Co-Investigator / Observer | Australian Astronomical Observatory, Siding Spring Observatory - *HERMES+2dF, Veloce*

Skills

Programming & Scientific Computing: Python (`NumPy`, `SciPy`, `Matplotlib`, `JAX`, `PyMC`, `galpy`)
Data Analysis & Statistical Methods: Bayesian Inference, MCMC, Gaussian Process Regression
Research Software & Tools: TOPCAT, Astropy, Git/GitHub

Teaching Experience

Academic Tutorials and Labs Supervised for undergraduate courses at Macquarie University:

- **PHYS1510 (Engineering Physics):** 2/2026, 2/2025, 09/2024, 02/2024, 09/2023, 02/2023
- **PHYS1210 (Physics for Life Sciences):** 2/2025; 09/2024, 02/2024, 09/2023, 02/2023
- **PHYS1020 (Electric and Magnetic Interactions):** 09/2024
- **ASTR3020 (Extragalactic Astronomy and Cosmology):** 09/2024

Hobbies and Interests

- Painting and sketching (freehand, watercolour)
- Playing musical instruments (guitar)
- Athletics: Javelin and Discus throw